Environment Setup

SQL Server and SSMS for the Data Analytics project, on Windows and Mac

The guide gets you to the first query of the project: SQL Server installed and the DataWarehouseAnalytics database in place, restored from the backup or built from the CSV files. With the fixes for the common errors.

1 Choose your setup

Path	For	Same screens as the videos	Cost
SQL Server Express + SSMS, section 3	Windows	yes	free
Docker, section 5	Mac	the same SQL, in Visual Studio Code	free
Parallels with Windows, section 6	Mac	yes, SSMS inside Windows	about 100 dollars a year, 14-day trial

Read only your path. Windows: sections 3 and 4. Mac: section 5, or section 6 for Windows inside the Mac. Everyone: section 7 when something breaks.

My advice: on Windows, take the first path, it is the one in the videos. On a Mac, take Docker: free, light, and the SQL is the same. Take Parallels if you want SSMS exactly as in the videos and your Mac has 16 GB of memory.

2 What you get in the materials

Download the project materials once. You get this folder:

```
sql-data-analytics-project/
├── 00_setup/        this guide, and the one-click Docker setup for Mac
├── datasets/
│   ├── DataWarehouseAnalytics.bak   the database, ready to restore
│   └── csv-files/   the same data as CSV, layer by layer: bronze, silver, gold
├── docs/            the project roadmap and the notes
└── scripts/         00_init_database.sql builds the database from the CSV files;
                     01 to 06 explore the data, 07 to 13 analyze it and build the two reports
```

The project runs on one database, `DataWarehouseAnalytics`. Its `gold` schema holds the three tables every query uses: `dim_customers`, `dim_products` and `fact_sales`. The scripts are the ones written in the videos, so you can type along or run them.

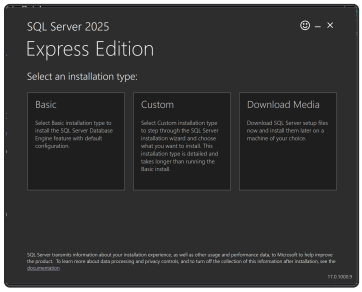
3 Windows: install SQL Server and SSMS

SQL Server Express is the free edition, and it is all you need. The videos show SQL Server 2022. The download page now offers 2025, and it works the same.

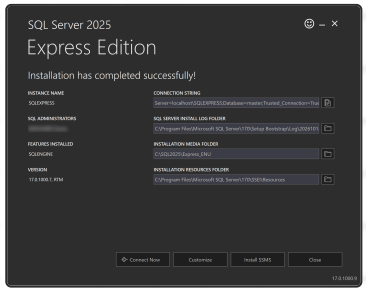
- Open the SQL Server downloads page, scroll to the **Express** card, and click **Download now**.
- Run the downloaded file and choose **Basic**. Accept the licence and click **Install**.
- The last screen shows your server name: `localhost\SQLEXPRESS`. Keep it, you need it in a minute. Your Windows account is now the administrator of this SQL Server.
- Download SQL Server Management Studio (SSMS) and install it with the default options. The installer may offer extra components. You need none of them.



Step 1: the Express card.



Step 2: choose Basic.



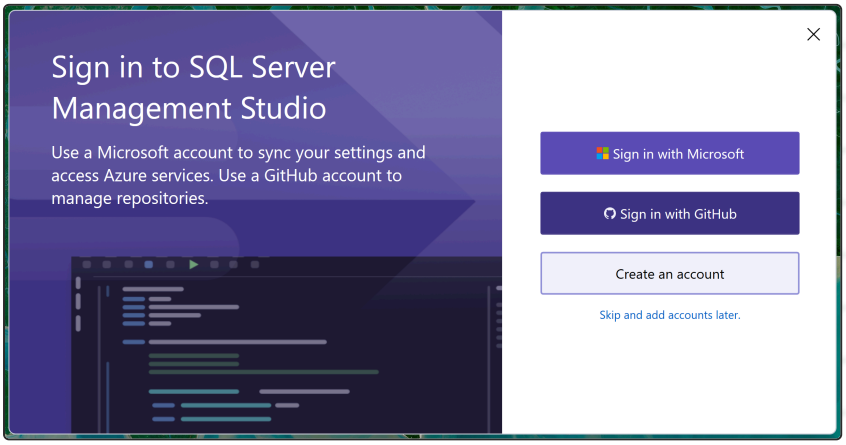
Step 3: the server name.

DATA WITH BARAA

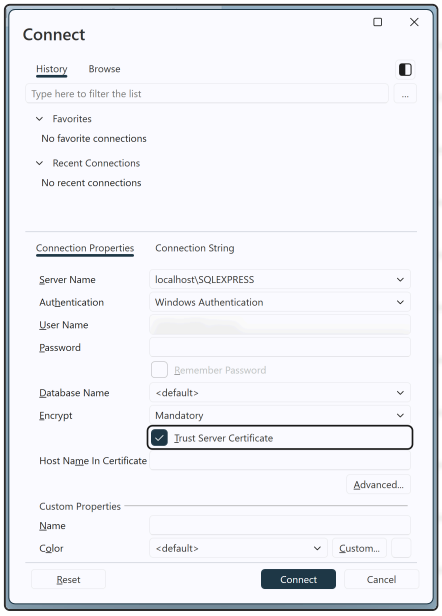
Connect for the first time

SSMS 21 and newer look different from the videos.

- At the first start, SSMS asks you to sign in. You do not need an account: click **Skip and add accounts later** under the buttons.
- In the **Connect** window, type the server name `localhost\SQLEXPRESS`. If the field is empty, that is normal.
- Authentication: **Windows Authentication**.
- Tick **Trust Server Certificate**. Without it you get a certificate error.
- Click **Connect**. Your server appears on the left, in the Object Explorer.



Step 1: the skip link under the buttons.



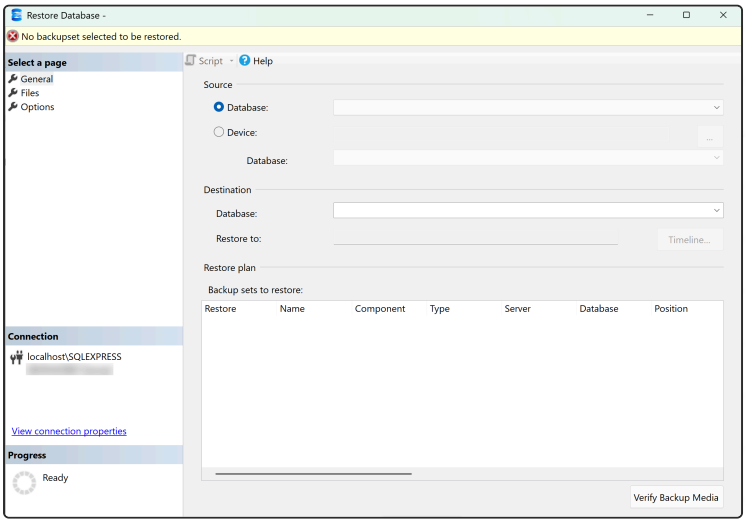
Steps 2 to 4: the Connect window, filled in.

4 Create the database

Two ways to the same database. The first is a few clicks and needs SQL Server 2022 or newer. The second builds it from the CSV files and works on every version.

Way 1: restore the backup

- Copy `datasets/DataWarehouseAnalytics.bak` into the Backup folder of your SQL Server: `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Backup` for SQL Server 2025, `MSSQL16` for 2022. SQL Server cannot read your Downloads folder, so this copy matters.
- In the Object Explorer, right-click **Databases > Restore Database**.
- Choose **Device**, click the three dots, click **Add**, pick the backup file, and click **OK** twice. Refresh the Databases folder: `DataWarehouseAnalytics` is there.



Way 1: Device, the three dots, Add.

> One rule to know: a backup restores on the same SQL Server version or a newer one, never on an older one. This backup comes from SQL Server 2022. On SQL Server 2019 or older, use way 2.

Way 2: build it from the CSV files

- Create the folder `C:\sql` and copy the whole project folder into it, so the files sit at `C:\sql\sql-data-analytics-project\datasets\csv-files\`. That is the path written in the script, the one you see in the videos. A folder directly on `C:\` is readable by SQL Server; Downloads and Desktop often are not.
- In SSMS, click **File > Open > File**, open `scripts/00_init_database.sql`, and press F5. It creates the database, the `gold` schema and the three tables, and loads them from the CSV files.

Then, for every script

The project scripts run inside `DataWarehouseAnalytics`. Before you execute, pick it in the database dropdown of the toolbar, where it says `master`. If you built the data warehouse project, you already have this gold layer in `DataWarehouse`, and the queries run there too.

5 Mac: Docker

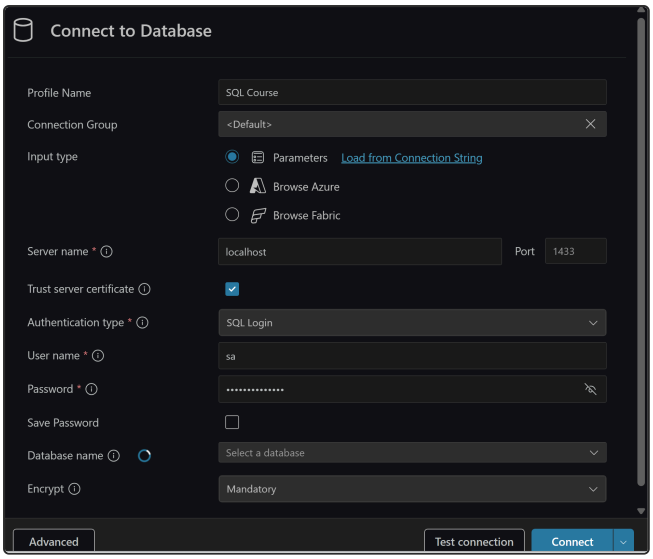
SQL Server does not install on macOS, and SSMS is Windows only. With Docker, SQL Server runs in a small container on your Mac, and you write your queries in Visual Studio Code. The SQL is the same.

Install, once

- Install Docker Desktop, the Apple chip version. Start it and leave it running.
- Docker Desktop settings, **General**. Set the Virtual Machine Manager to **Apple Virtualization framework** and turn on **Use Rosetta for x86_64/amd64 emulation on Apple Silicon**. SQL Server ships for Intel chips only, and this setting lets your Mac run it.
- Install Visual Studio Code and, inside it, the extension SQL Server (mssql) by Microsoft. It is Microsoft's tool for the Mac.

Start the project database: one click

- In the project materials, open `00_setup/mac-docker` and double-click `start.command`.
- Wait a minute. The first start downloads SQL Server. Then the command puts the project files inside SQL Server at `C:\sql\sql-data-analytics-project\` and runs the project's own `00_init_database.sql`, which builds `DataWarehouseAnalytics` from the CSV files. SQL Server on Linux reads `C:\` as its root folder, so the script runs exactly as in the videos.
- In Visual Studio Code, open the SQL Server extension on the left and add a connection. Server name `localhost`, port `1433`, tick **Trust server certificate**, authentication **SQL Login**, user `sa`, password `SqlServer!2026`.
- Switch the database to `DataWarehouseAnalytics`: click `master` in the bar at the bottom of the window and pick it. Open `scripts/01_database_exploration.sql`, run it, and you are in the project.



Step 3: the connection form, filled in.

The same by hand

If you prefer your own commands: start SQL Server, copy the project in, then run `00_init_database.sql` from Visual Studio Code. Choose your own password: at least 8 characters with upper case, lower case, a number and a symbol.

```
docker run -e "ACCEPT_EULA=Y" -e "MSSQL_SA_PASSWORD=YourStrong!Passw0rd" \
  -p 1433:1433 --name sql --platform linux/amd64 \
  -d mcr.microsoft.com/mssql/server:2022-latest
docker exec -u root sql mkdir -p /sql/sql-data-analytics-project
docker cp path/to/sql-data-analytics-project/datasets sql:/sql/sql-data-analytics-project/
docker exec -u root sql chown -R mssql /sql
```

Microsoft's own guide for the first command: SQL Server in a Docker container.

> Did you run the SQL course package or the data warehouse package on this Mac? Its SQL Server is already there: the project's `start.command` finds the running container and uses it, with the same connection.

6 Windows inside your Mac: Parallels

This path gives you SSMS exactly as in the videos, because you run Windows on your Mac.

- Install Parallels Desktop. It downloads and installs Windows 11 for you in a few clicks. It costs about 100 dollars a year, with a 14-day free trial, and Windows needs its own licence.
- Give the virtual machine at least 6 GB of memory. A Mac with 16 GB is the comfortable minimum.
- Inside Windows, follow sections 3 and 4 like any Windows user.

SQL Server Express installs and runs on the ARM version of Windows 11, even though Microsoft does not list it as supported. Tested with SQL Server 2025 Express and SSMS 22 in Parallels on an Apple chip.

7 When something breaks

Find your message in the first column. The fix is in the last one.

Install and connect

What you see	Why	What to do
SSMS asks you to sign in at the first start	optional account	Click **Skip and add accounts later**
The server name field is empty	normal	Type `localhost\SQLEXPRESS`
Error 26, "Error Locating Server/Instance Specified"	SQL Server is not running, or the name is wrong	Open **Services**, start "SQL Server (SQLEXPRESS)". If you installed the Developer edition, the name is `localhost` without the instance
"The certificate chain was issued by an authority that is not trusted"	SSMS encrypts the connection	Tick **Trust Server Certificate**
Installer: "Wait on the Database Engine recovery handle failed"	SQL Server could not start during setup	Uninstall SQL Server, restart Windows, install again. If it fails again, the real reason is in the ERRORLOG file under `C:\Program Files\Microsoft SQL Server\MSSQL17.SQLEXPRESS\MSSQL\Log`
SQL Server 2025 setup stops at the rules check	installer problem on some machines	Install SQL Server 2022 Express instead. Everything here runs on it
Your laptop does not let you install anything	no administrator rights	SQL Server needs them. Ask IT, or use a personal computer

The database and the files

What you see	Why	What to do
The restore window does not show the .bak file, or "Access is denied"	SQL Server cannot read Downloads	Copy the file into the Backup folder, section 4, way 1
Msg 3169, the backup "is incompatible with this server"	your SQL Server is older than 2022	Build the database from the CSV files, way 2, or install SQL Server 2022 or 2025 Express
Msg 262, "CREATE DATABASE permission denied in database 'master'"	your Windows login is not an administrator of this SQL Server. Someone else installed it, often IT	Log in with the Windows account that installed SQL Server, or ask an administrator to add your login to the **sysadmin** role (Security > Logins > Properties > Server Roles)
Msg 4860, "Cannot bulk load. The file does not exist or you don't have file access rights"	the file is not at the path in the script	The script reads `C:\sql\sql-data-analytics-project\datasets\csv-files\` . Put the folder there, section 4, or change the three paths in `00_init_database.sql`
Msg 4861, "Operating system error code 5 (Access is denied.)"	SQL Server runs as a service and cannot read your folder	Move the folder to `C:\sql\` . Downloads, Desktop and OneDrive folders are the usual culprits
Msg 208, "Invalid object name 'gold.fact_sales'"	the query runs in `master` , not in the project database	Pick `DataWarehouseAnalytics` in the database dropdown and run again

What you see	Why	What to do
"'DATETRUNC' is not a recognized built-in function name"	SQL Server older than 2022	Install SQL Server 2022 or newer. The change-over-time scripts use functions from 2022
A .sql file opens in Notepad, or the .bak shows as text in the browser	the file is not linked to SSMS	Open SSMS first, then **File > Open > File**. For downloads, right-click the link and choose **Save link as**

Docker

What you see	Why	What to do
"Cannot connect to the Docker daemon"	Docker Desktop is not running	Start Docker Desktop, wait for the whale icon, try again
"no matching manifest for linux/arm64"	the Intel image needs emulation	Turn on Rosetta in Docker Desktop, section 5. By hand, add `--platform linux/amd64` to the command
"port is already allocated"	another SQL Server listens on 1433, not one of these packages	Stop it, or change the first `1433` in `docker-compose.yml` to `1434` and connect to port `1434`
The container stops right after the start	the password breaks the rules	At least 8 characters with upper case, lower case, a number and a symbol
"SQL Server did not answer" from start.command	SQL Server could not start, most often Rosetta is off	Docker Desktop settings, **General**, turn on Rosetta, section 5. The container's log in Docker Desktop names the reason

If you are still stuck, write in the Q&A with the exact error message and a screenshot of the window. Then I can see what happened.